HYPOTHESIS TESTING

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 13

5-MINUTE REVISION

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1. QUICK REVISION · 1

- Inferential statistics → uses sample data to draw conclusions about a population.
- Hypothesis → tentative and testable statement.
- Statistical hypothesis → testable statement about a population parameter.
- H_0 → null hypothesis; generally represents no effect, no difference, or no relationship.
- H_1 or H_a → alternative hypothesis; represents an effect, difference, or relationship.

2. QUICK REVISION · 2

- Falsification principle → a scientific hypothesis should be capable of being tested and potentially falsified.
- Karl Popper → associated with the falsification principle.
- Always use "fail to reject H_0"; do not say "accept H_0".
- Test statistic → numerical function of sample observations used to make a statistical decision.
- Common test statistics → Z, t, χ^2, and F.

3. QUICK REVISION · 3

- Type I error → reject H_0 when H_0 is true.
- Type II error → fail to reject H_0 when H_0 is false.
- α → chosen upper bound on the Type I error probability.
- β → probability of Type II error.
- Power → probability of correctly rejecting a false H_0.

4. QUICK REVISION · 4

- Power → $1 - \beta$.
- Larger sample sizes generally increase power.
- Sampling distribution → probability distribution of possible values of a test statistic under repeated sampling, assuming H_0 is true.
- Rejection region → values of the test statistic for which H_0 is rejected.
- Critical region → another name for rejection region.

5. QUICK REVISION · 5

- Acceptance region → complementary region where H_0 is not rejected.
- Critical value → value separating the rejection and non-rejection regions.
- Under H_0, the rejection probability is controlled at α; a discrete test can be conservative.
- Under H_0, the non-rejection probability is at least $1 - \alpha$ for a level-α test.
- $H_1 : \mu < \mu_0$ → left-tailed test.

6. QUICK REVISION · 6

- $H_1 : \mu > \mu_0$ → right-tailed test.
- $H_1 : \mu \neq \mu_0$ → two-tailed test.
- An equal-tail two-sided test allocates $\alpha/2$ to each tail; a discrete test may not attain these probabilities exactly.
- Right-tailed test → reject H_0 when the calculated statistic is greater than the critical value.
- Left-tailed test → reject H_0 when the calculated statistic is less than the critical value.

7. QUICK REVISION · 7

- Two-tailed test → reject H_0 when the statistic falls below the lower critical value or above the upper critical value.
- p-value → probability of obtaining a test statistic as extreme as or more extreme than the observed value, assuming H_0 is true.
- Small p-value → strong evidence against H_0.
- Large p-value → insufficient evidence to reject H_0.
- p-value $\leq \alpha$ → reject H_0.

8. QUICK REVISION · 8

- p-value $> \alpha$ → do not reject H_0.
- Failing to reject H_0 does not prove that H_0 is true.
- The alternative hypothesis determines whether the test is one-tailed or two-tailed.
- The level of significance is selected before conducting the test.
- Common significance levels in the chapter → $\alpha = 0.05$ and $\alpha = 0.01$.

9. QUICK REVISION · 9

- The eight basic steps are: state H_0, state H_1, set α, collect data, calculate the test statistic, identify the critical value or region, compare, and make the decision.
- A statistical decision can be incorrect because it is based on sample data.
- Type I and Type II errors have different consequences depending on the context.
- Fisher's tea-tasting experiment is a historical illustration of hypothesis testing and significance testing.

10. IMPORTANT FORMULAE · 1

Level of significance:

$$P(\text{Reject } H_0 \mid H_0 \text{ is true}) \leq \alpha$$

Probability of Type II error:

$$\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

Power of a test:

$$\text{Power} = 1 - \beta$$

11. IMPORTANT FORMULAE · 2

Size of rejection region:

$$P_{H_0}(\text{Rejection region}) \leq \alpha$$

Size of acceptance region:

$$P_{H_0}(\text{Non-rejection region}) \geq 1 - \alpha$$

Binomial test statistic used in the worked coin example:

$$\hat{p} = \frac{x}{n}$$

12. IMPORTANT FORMULAE · 3

Binomial probability:

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

Equivalent binomial probability formula:

$$P(X = x) = \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

Decision rule based on p-value:

$$\text{Reject } H_0 \quad \text{if} \quad p\text{-value} \leq \alpha$$

13. IMPORTANT FORMULAE · 4

Non-rejection rule based on p-value:

$$\text{Do not reject } H_0 \quad \text{if} \quad p\text{-value} > \alpha$$

14. SELF-CHECK FROM THE CHAPTER

1. A testable statement about a population distribution or its parameters is called a statistical _____. **[Hypothesis]**
2. The hypothesis tested for possible rejection is called the _____ hypothesis. **[Null]**
3. The hypothesis specifying a departure from the null hypothesis is called the _____ hypothesis. **[Alternative]**
4. Rejecting a true null hypothesis is a Type _____ error. **[I]**